

Computer Programming Test

The Appliance class implements the Valuable and Comparable interfaces and has the following characteristics:

- o Its attributes are double priceBase, String color, char letter and int weight.
- o By default, the color will be white and the letter will be 'F'.
- o A constructor for attributes with non-predefined value.
- o Methods get and set for all attributes.

The Valuable interface only has the double method calculatePrice (), which will calculate and return the total price of each appliance.

To calculate the price, the letter, weight and color will be taken into account as follows: (A, €100), (B, €90), (C, €70), (D, €50), (E, €40) and (F, €20). Weight more than 80kg €100, (70-80kg, 80), (60-70kg, 60), (40-60kg, 50) and (less than 40kg, €40). Any color other than "White" has an increase of €30. The sum of these values will be added to the base price which will be the price for a generic appliance.

Score: 2 points

The Washer class inherits from Appliance and has the following characteristics:

- o An int load attribute.
- o A constructor with the load and the rest of inherited attributes.
- o The necessary get and set methods and a toString () method to show their values including its total price.
- o If it has a load greater than 10kg, its price will increase by €50.

The Television class inherits from Appliance and has the following characteristics:

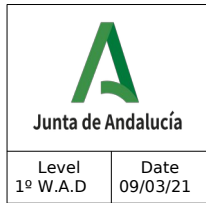
- o Its attributes are int length (in inches) and boolean WIFI.
- o By default the WIFI will be false and its color will be black.
- o A constructor with the length and the rest of inherited attributes.
- o The necessary get and set methods and a toString () method to show their values, including its total price.
- o If it has WIFI, the price will be increased by €40, and if it has a length greater than 40 inches, the price will increase by 25% after calculated all the possible increases.

Score: 2 points

The Main class creates an array of 2 Washers and another of 2 Televisions.

The first washing machine has a price of €100, it is white, it weighs 80kg, it has a load of 12kg and it is letter B.

The second washing machine has a price of €150, it is gray, it weighs 50kg, it has an 8kg load and it is letter F.



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The first TV has a price of €50, is black, has WIFI, letter 'B', 30 inches and weight 10kg.
The second TV has a price of €200, is gray, letter A, 50 inches and weighs 65kg.

Loop through both arrays to show all the appliances. **(1 point)**

Create an array of 4 appliances from the arrays of washing machines and TV. **(1 point)**

Order the array of appliances for the total price of each one and show it from highest to lowest price. **(1 point)**

Show all the data of the washing machines, from lowest to highest price. **(1 point)**

Ask for a color by keyboard and show all the data of the appliances of that color. **(1 point)**

Change the base price of all the appliances. If the appliance is a washing machine, add 20 euros to its base price and if it is a TV, subtract 10 euros from its base price. Finally, loop through the array of appliances to show the changes. **(1 point)**

Score: 6 points

Annotations:

Time 2:00 hours. The name of the project will be your surname plus your first name. Export the project in the same way. Example: MoralesLucenaMaria.zip.

Runtime and compiling errors will not be accepted in the test. Students cannot assume anything without consulting or asking teachers.